

Library Demand Forecaster & Inter-Branch Inventory Rebalancing Engine

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HACKATHON TRACK - Track A - DataDrishti (Power BI / Tableau)

1. Introduction

Libraries in universities and large institutions manage thousands of books across multiple branches. Understanding borrowing patterns, identifying high-demand titles, and maintaining the right inventory at each branch are essential for providing efficient services to students, faculty, researchers, and staff.

The **Library Demand Forecaster & Inter-Branch Inventory Rebalancing Engine** is a Power BI-based analytical solution that transforms library transaction data into actionable insights. The system analyzes historical borrowing patterns, patron behavior, overdue transactions, category-level demand, and branch-level stock shortages.

The dashboard also uses time-series forecasting to estimate future demand and helps library administrators identify opportunities for transferring books between branches.

2. Problem Statement

University and institutional libraries frequently face difficulties in maintaining the right books at the right branches.

The major problems identified are:

2.1 Unpredictable Demand

Book demand changes significantly depending on academic activities such as:

- Examinations
- Assignments
- Project submissions
- Semester schedules
- Research activities

High-demand books can therefore become unavailable unexpectedly.

2.2 Branch-Level Inventory Imbalance

Some branches may experience book shortages while other branches have unused copies of the same or similar titles.

This creates an inefficient distribution of physical inventory.

2.3 Lack of Demand Forecasting

Traditional library systems generally focus on recording transactions rather than predicting future demand.

Without forecasting, administrators may not know:

- Which categories will become highly demanded
- When borrowing demand may increase
- Which books should be stocked in advance

2.4 High-Demand Book Shortages

Popular titles in areas such as:

- Artificial Intelligence
- Machine Learning
- Data Science
- Programming
- Software Engineering

can experience high borrowing activity and stockout events.

2.5 Limited Analytical Visibility

Library administrators may need to manually analyze transaction records to understand:

- Borrowing trends
- User behavior
- Overdue patterns
- Branch shortages
- Category demand
- Popular titles

This makes decision-making slower and less data-driven.

3. Existing System

The existing library management approach primarily focuses on **transaction recording and basic inventory management**.

Typical operations include:

- Book issue and return
- Member registration
- Book availability checking
- Reservation management
- Overdue tracking
- Inventory recording

However, the existing approach has several limitations.

Limitations of the Existing System

Existing Capability	Limitation
Book issue/return tracking	Mainly records historical transactions
Inventory management	Does not effectively identify redistribution opportunities
Reservation tracking	Shows current reservations but provides limited predictive insight
Overdue tracking	Does not provide deeper behavioral analysis
Branch management	Limited cross-branch comparison
Historical reports	Mostly descriptive rather than predictive
Demand analysis	Requires manual analysis
Future demand planning	Limited forecasting capability

Main Gap

The major limitation is the absence of an integrated analytical layer that combines **historical circulation, user behavior, demand forecasting, and branch inventory shortages**.

Therefore, library administrators may react to shortages only after they occur rather than taking preventive action.

4. Proposed Solution

The proposed solution is a **Power BI analytical dashboard** that integrates library transaction data and converts it into actionable business intelligence.

The system provides four major capabilities:

4.1 Demand Analysis

Historical borrowing transactions are analyzed to identify:

- Daily/weekly borrowing patterns
- Category demand
- Popular books
- Branch-level circulation

4.2 Predictive Demand Forecasting

Power BI time-series forecasting is used to estimate future checkout demand.

The forecast includes a **95% confidence interval**, allowing administrators to understand both expected demand and possible variation.

4.3 Patron Segmentation

Borrowers are analyzed according to their roles:

- Students
- Faculty
- Research Scholars
- Staff

This helps administrators understand which user groups generate demand at different branches.

4.4 Inter-Branch Inventory Rebalancing

Zero-stock transactions are identified using branch and book-level analysis.

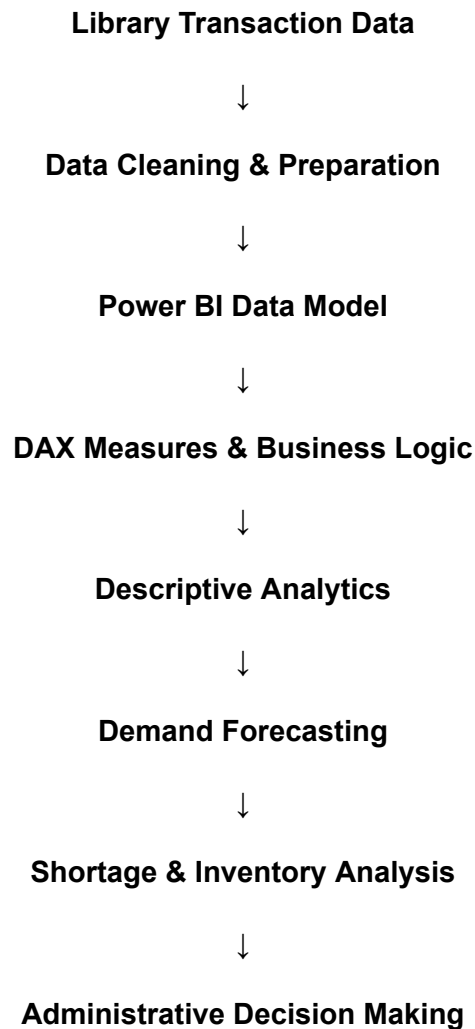
This allows administrators to identify situations where:

One branch experiences shortages while another branch may potentially have available inventory.

The information can support physical book transfers and inventory redistribution.

5. System Architecture

The analytical workflow can be represented as:



The solution therefore transforms raw transaction records into a decision-support system for library management.

6. Dataset Description

The project uses the dataset:

library_data_updated.csv

The dataset contains **1,000 library transaction records**.

Important Fields

Field	Description
Transaction_ID	Unique transaction identifier
User_ID	Unique patron identifier
User_Role	Student, Faculty, Research Scholar, or Staff
Book_ID	Unique book identifier
Book_Title	Title of the borrowed book
Book_Category	Subject/category of the book
Branch_Location	Library branch
Borrow_Date	Date on which the book was borrowed
Return_Date	Date on which the book was returned
Borrow_Duration_Days	Number of days borrowed
Overdue_Status	Indicates whether the transaction was overdue
Renewal_Count	Number of renewals
Available_Copies_At_Borrow	Copies available at checkout
Reservation_Status	Reservation/hold status

7. Dashboard Implementation

Page 1 — Executive Demand & Circulation Dashboard

The first dashboard provides an executive-level overview of library activity.

Key Performance Indicators

The dashboard displays:

- **Total Checkouts:** 1,000
- **Active Borrowers:** 78
- **Overdue Rate:** 10.30%
- **Zero Stock Transactions:** 98

These KPIs provide an immediate understanding of the overall library situation.

Demand Forecast

The demand forecast visual compares historical checkout volume against predicted future demand.

A **90-day forecast** is generated with a **95% confidence interval**, helping administrators anticipate potential increases in borrowing activity.

User Role Analysis

A 100% stacked bar chart compares user roles across branches.

This identifies the dominant borrowing groups at each facility and helps administrators understand branch-specific patron behavior.

Branch Shortage Matrix

A book-versus-branch matrix identifies zero-stock transactions.

This acts as a shortage detection mechanism and highlights titles that may require redistribution or additional procurement.

Screenshot — Page 1

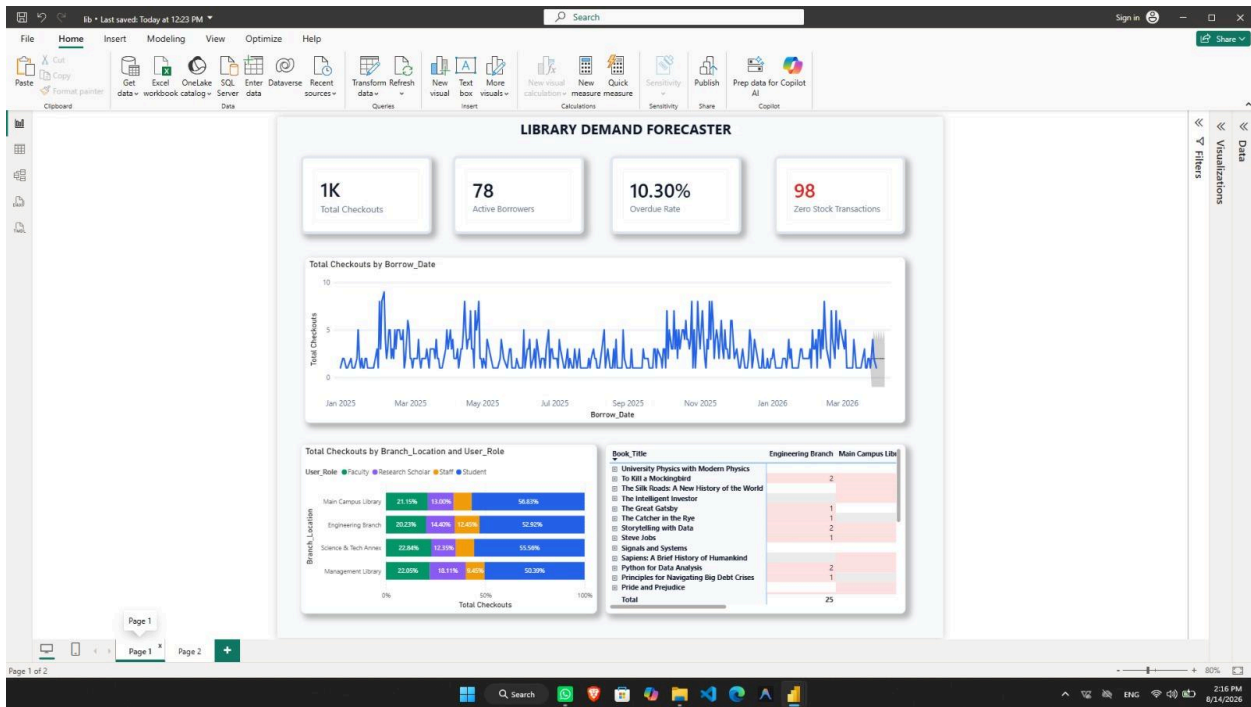


Figure 1: Executive Demand & Circulation Dashboard

8. Page 2 — Genre Analytics & Catalog Drill-Down

The second dashboard focuses on detailed category and catalog analysis.

Category Demand Share

The category distribution shows:

Category	Checkouts	Share
Computer Science & AI	271	27.1%
Data Science & Statistics	220	22.0%

Fiction & Literature	167	16.7%
History & Biography	164	16.4%
Business & Economics	108	10.8%
Science & Engineering	70	7.0%

The results indicate that **Computer Science & AI** has the highest borrowing demand.

Top Borrowed Books

The eight most borrowed titles are:

1. *Hands-On Machine Learning* — 54
2. *Storytelling with Data* — 49
3. *Artificial Intelligence: A Modern Approach* — 48
4. *Deep Learning with Python* — 48
5. *Clean Code: A Handbook of Agile Software* — 47
6. *Designing Data-Intensive Applications* — 45
7. *Operating System Concepts* — 44
8. *Python for Data Analysis* — 44

These results can help library administrators prioritize procurement and inventory redistribution.

Recent Category Demand

A stacked area chart displays category-level demand trends over the recent 90-day period.

This helps identify periods where specific academic domains experience increased borrowing.

Screenshot — Page 2

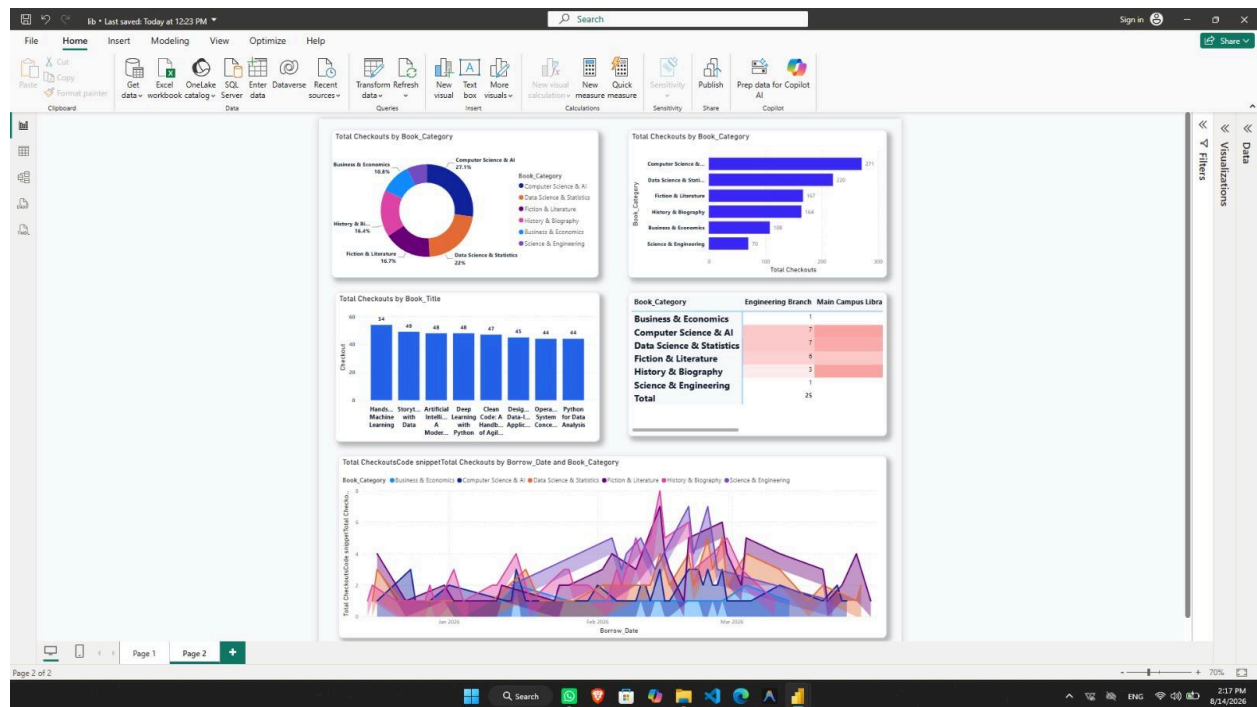


Figure 2: Genre Analytics & Catalog Drill-Down Dashboard

9. DAX Measures and Business Logic

The dashboard uses DAX measures to calculate important business metrics.

Total Checkouts

Total Checkouts =

COUNT('library_data_updated'[Transaction_ID])

This measure calculates the total number of borrowing transactions.

Active Borrowers

Active Borrowers =

DISTINCTCOUNT('library_data_updated'[User_ID])

This calculates the number of unique library users.

Overdue Rate

Overdue Rate =

```
DIVIDE(  
  CALCULATE(  
    COUNTROWS('library_data_updated'),  
    'library_data_updated'[Overdue_Status] = 1  
  ),  
  COUNTROWS('library_data_updated'),  
  0  
)
```

This calculates the percentage of transactions that exceeded the permitted borrowing duration.

Zero Stock Transactions

Zero Stock Transactions =

```
CALCULATE(  
  [Total Checkouts],  
  'library_data_updated'[Available_Copies_At_Borrow] = 0  
)
```

This identifies transactions where no physical copy was available at the time of borrowing.

Alert Color

Alert Color Hex =

```
IF(  
  [Zero Stock Transactions] > 15,  
  "#DC2626",  
  "#059669"  
)
```

This provides a conditional alert mechanism:

- **Red:** High shortage risk
- **Green:** Lower shortage level

10. Key Analytical Insights

The dashboard provides several important insights.

Insight 1 — Computer Science & AI Has the Highest Demand

Computer Science & AI accounts for **271 checkouts**, representing **27.1%** of total borrowing activity.

This indicates strong demand for technical and AI-related resources.

Insight 2 — Data Science Is the Second-Highest Demand Category

Data Science & Statistics contributes **220 checkouts**, representing **22.0%** of total activity.

Together, Computer Science & AI and Data Science & Statistics represent nearly half of all recorded transactions.

Insight 3 — Popular Technical Titles Require Priority Management

Hands-On Machine Learning is the most borrowed title with **54 checkouts**.

Other highly demanded technical books include:

- *Artificial Intelligence: A Modern Approach*
- *Deep Learning with Python*
- *Clean Code*
- *Designing Data-Intensive Applications*
- *Python for Data Analysis*

These titles should receive priority during stock planning.

Insight 4 — Stockout Events Require Attention

The dashboard identifies **98 zero-stock transactions**.

This indicates that users attempted to borrow books when no physical copies were available.

Such events can increase waiting time and reduce user satisfaction.

Insight 5 — Branch-Level Demand Is Different

The user-role distribution varies between branches.

For example, the Main Campus Library has a strong student borrowing share, while other branches show different combinations of faculty, research scholars, and staff.

Therefore, inventory planning should be performed at the **branch level rather than using a single institution-wide inventory strategy**.

11. Benefits of the Proposed Solution

The Power BI solution provides the following benefits:

For Library Administrators

- Faster decision-making
- Centralized performance monitoring
- Better inventory planning
- Early identification of shortages
- Data-driven procurement decisions

For Library Users

- Reduced waiting time
- Better availability of popular books
- Improved access to high-demand resources
- Reduced probability of unnecessary reservations

For Library Operations

- Better utilization of existing inventory
 - Identification of branch-level imbalances
 - Improved demand planning
 - Reduced manual reporting effort
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12. Conclusion

The **Library Demand Forecaster & Inter-Branch Inventory Rebalancing Engine** demonstrates how Power BI can transform traditional library transaction data into a practical decision-support system.

The solution combines **descriptive analytics, patron segmentation, time-series forecasting, category analysis, and stockout detection** within a single dashboard.

By identifying high-demand resources and branch-level shortages, the system enables library administrators to make proactive decisions regarding **inventory redistribution, procurement, and demand planning**.

The proposed solution can therefore help institutional libraries move from a **reactive inventory management model to a proactive, data-driven management approach**.

13. Future Enhancements

The solution can be further enhanced by implementing:

- Automated inter-branch transfer recommendations
 - Machine-learning-based demand forecasting
 - Real-time inventory integration
 - Email/notification alerts for critical shortages
 - Book-level reorder recommendations
 - Reservation and waitlist prediction
 - Integration with existing Library Management Systems
 - Automated monthly management reports
 - What-if analysis for inventory redistribution
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